

# EE3621: Digital Signal Processing

## Student Lecture Notes — Lecture 22 Design of FIR Filters Using the Windowing Method

### III B.Tech EEE — Core Exam Preparation Guide

---

## 1 Core Concept & Principle of Windowing

### 1.1 The Truncation Problem & Ideal Impulse Response Derivation

An ideal brick-wall lowpass filter with cutoff frequency  $\omega_c$  has the desired frequency response:

$$H_d(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases} \quad (1)$$

The continuous impulse response  $h_d[n]$  is obtained via the **\*\*Inverse DTFT\*\***:

$$\begin{aligned} h_{d,\text{LP}}[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \left[ \frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c} = \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{2\pi jn} \end{aligned} \quad (2)$$

Applying Euler's sine formula  $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ :

$$h_{d,\text{LP}}[n] = \frac{2j \sin(\omega_c n)}{2\pi jn} = \frac{\sin(\omega_c n)}{\pi n}, \quad -\infty < n < \infty \quad (3)$$

At  $n = 0$ , applying L'Hôpital's rule:  $h_{d,\text{LP}}[0] = \lim_{n \rightarrow 0} \frac{\sin(\omega_c n)}{\pi n} = \frac{\omega_c}{\pi}$ . This impulse response is doubly unphysical: it has **\*\*infinite duration (IIR)\*\*** and is **\*\*non-causal\*\*** (extends to  $-\infty$ ).

To realize a physical causal FIR filter of length  $N$ , we:

1. Shift the center of symmetry to  $\tau = \alpha = \frac{N-1}{2}$  to ensure causality and linear phase.
2. Multiply  $h_d[n]$  pointwise by a finite-duration window sequence  $w[n]$  ( $0 \leq n \leq N-1$ ):

$$\mathbf{h}[\mathbf{n}] = \mathbf{h}_d[\mathbf{n}] \cdot \mathbf{w}[\mathbf{n}], \quad 0 \leq n \leq N-1 \quad (4)$$

### 1.2 Frequency-Domain Convolution & Gibbs Phenomenon

By the DTFT modulation property, time-domain multiplication corresponds to periodic convolution in frequency:

$$H(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) \circledast W(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta \quad (5)$$

- **Mainlobe Width of  $W(e^{j\omega})$ :** Spreads out the ideal vertical transition into a gradual transition band  $\Delta\omega$ .
- **Sidelobes of  $W(e^{j\omega})$ :** Cause oscillations (ringing) in the passband and stopband.
- **Gibbs Phenomenon:** Abrupt rectangular truncation results in a fixed 8.95% overshoot ( $\approx -21$  dB peak sidelobe) near the discontinuity, which **cannot be reduced by increasing  $N$** . Smoother, tapered windows are necessary to suppress sidelobes.

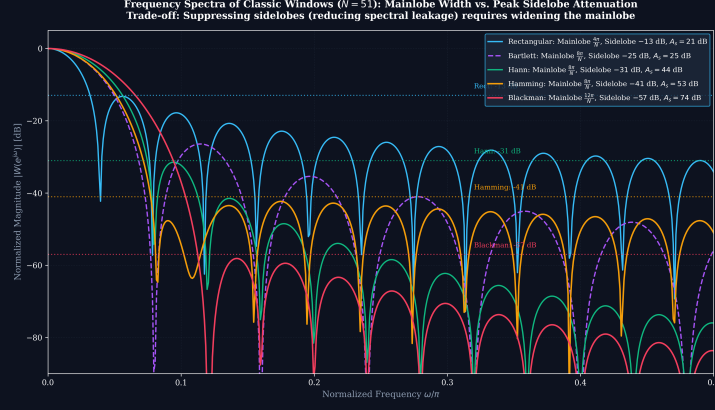


Figure 1: Frequency spectra of classic window functions showing the fundamental trade-off: narrower mainlobe (faster transition) vs. lower sidelobes (higher stopband attenuation).

## 2 Master Window Summary Table (Formulas & Specs)

Table 1: Characteristics of Standard FIR Filter Window Functions

| Window Name           | Time-Domain Equation $w[n]$ ( $0 \leq n \leq N-1$ )                                                | Transition Width $\Delta\omega$           | Peak Sidelobe | Min. Attenuation $A_s$ |
|-----------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------|---------------|------------------------|
| Rectangular           | $w[n] = 1$                                                                                         | $\frac{4\pi}{N}$ (or $\frac{1.8\pi}{N}$ ) | -13 dB        | 21 dB                  |
| Bartlett (Triangular) | $w[n] = 1 - \frac{2 n-(N-1)/2 }{N-1}$                                                              | $\frac{8\pi}{N}$ (or $\frac{6.1\pi}{N}$ ) | -25 dB        | 25 dB                  |
| Hann (Hanning)        | $w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right)$                                             | $\frac{8\pi}{N}$ (or $\frac{6.2\pi}{N}$ ) | -32 dB        | 44 dB                  |
| Hamming               | $w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$                                           | $\frac{8\pi}{N}$ (or $\frac{6.6\pi}{N}$ ) | -43 dB        | <b>53 dB</b>           |
| Blackman              | $w[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$ | $\frac{12\pi}{N}$ (or $\frac{11\pi}{N}$ ) | -58 dB        | <b>74 dB</b>           |

## 3 Ideal Shifted Impulse Response Formulations

To design a causal, linear-phase FIR filter of length  $N$ , the ideal frequency response incorporates a linear phase delay  $\tau = \alpha = \frac{N-1}{2}$ :

$$H_d(e^{j\omega}) = \begin{cases} 1 \cdot e^{-j\omega\tau}, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases} \quad (6)$$

Evaluating the Inverse DTFT with all intermediate integral steps:

$$\begin{aligned} h_d[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\tau} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\tau)} d\omega = \frac{1}{2\pi} \left[ \frac{e^{j\omega(n-\tau)}}{j(n-\tau)} \right]_{-\omega_c}^{\omega_c} \\ &= \frac{e^{j\omega_c(n-\tau)} - e^{-j\omega_c(n-\tau)}}{2\pi j(n-\tau)} = \frac{2j \sin(\omega_c(n-\tau))}{2\pi j(n-\tau)} = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}, \quad n \neq \tau \end{aligned} \quad (7)$$

At the center of symmetry  $n = \tau$ , applying L'Hôpital's rule:

$$h_d[\tau] = \lim_{n \rightarrow \tau} \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)} = \lim_{n \rightarrow \tau} \frac{\omega_c \cos(\omega_c(n-\tau))}{\pi \cdot 1} = \frac{\omega_c}{\pi} \quad (8)$$

Highpass, bandpass, and bandstop filters follow directly from linear combinations of ideal lowpass filters:

- **Highpass:** By spectral inversion,  $H_{d,\text{HP}}(e^{j\omega}) = e^{-j\omega\tau} - H_{d,\text{LP}}(e^{j\omega}) \implies h_{d,\text{HP}}[n] = \delta[n - \tau] - h_{d,\text{LP}}[n]$ .
- **Bandpass:** As difference of two lowpass filters,  $h_{d,\text{BP}}[n] = h_{d,\text{LP2}}[n] - h_{d,\text{LP1}}[n]$ .
- **Bandstop:**  $h_{d,\text{BS}}[n] = \delta[n - \tau] - h_{d,\text{BP}}[n]$ .

Table 2: Closed-Form Formulas for Ideal Shifted Filters  $h_d[n]$

| Filter Type    | For $n \neq \tau$                                                                    | Center Tap $n = \tau$                                   |
|----------------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| Lowpass (LPF)  | $h_d[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$                                | $h_d[\tau] = \frac{\omega_c}{\pi}$                      |
| Highpass (HPF) | $h_d[n] = -\frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$                               | $h_d[\tau] = 1 - \frac{\omega_c}{\pi}$                  |
| Bandpass (BPF) | $h_d[n] = \frac{\sin(\omega_{c2}(n-\tau)) - \sin(\omega_{c1}(n-\tau))}{\pi(n-\tau)}$ | $h_d[\tau] = \frac{\omega_{c2} - \omega_{c1}}{\pi}$     |
| Bandstop (BSF) | $h_d[n] = \frac{\sin(\omega_{c1}(n-\tau)) - \sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)}$ | $h_d[\tau] = 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi}$ |

## 4 5-Step FIR Filter Window Design Procedure

1. **Identify Required Attenuation  $A_s$ :** From stopband specification  $A_s = -20 \log_{10}(\delta_s)$  dB, select the appropriate window from Table 1 that meets or exceeds  $A_s$ .
2. **Calculate Filter Length  $N$ :** From the transition bandwidth  $\Delta\omega = |\omega_s - \omega_p|$ , calculate  $N$  using the window's transition width formula. Round up to the **next odd integer** to ensure Type 1 linear phase.
3. **Determine Cutoff Frequency  $\omega_c$ :** Average the passband and stopband edge frequencies:

$$\omega_c = \frac{\omega_p + \omega_s}{2} \quad (9)$$

4. **Compute Ideal Impulse Response  $h_d[n]$ :** Evaluate  $h_d[n]$  for  $n = 0, 1, \dots, N - 1$  using Table 2.
5. **Apply Window:** Compute final filter coefficients:  $h[n] = h_d[n] \cdot w[n]$ .

## 5 Standard Solved Exam Example

**Problem:** Design a linear-phase FIR lowpass filter with a cutoff frequency  $\omega_c = 0.5\pi$  rad/sample and filter length  $N = 7$  using a **Hamming window**. Determine the impulse response  $h[n]$ , the system function  $H(z)$ , and evaluate the frequency response magnitude at  $\omega = 0$  (DC) and  $\omega = \pi$  (Nyquist).

**Solution:**

**Step 1: Symmetry Delay Parameter** For length  $N = 7$  (odd), the center of symmetry is:

$$\tau = \alpha = \frac{N - 1}{2} = \frac{7 - 1}{2} = 3 \quad (10)$$

**Step 2: Ideal Impulse Response Evaluation  $h_d[n]$**  The ideal shifted lowpass filter formula with  $\omega_c = 0.5\pi$  is:

$$h_d[n] = \begin{cases} \frac{\sin(0.5\pi(n-3))}{\pi(n-3)}, & n \neq 3 \\ \frac{\omega_c}{\pi} = \frac{0.5\pi}{\pi} = 0.5, & n = 3 \end{cases} \quad (11)$$

Computing each value explicitly for  $n = 3, 2, 1, 0$ :

$$\begin{aligned}
n = 3 : \quad h_d[3] &= 0.5000 \\
n = 2 : \quad h_d[2] &= \frac{\sin(0.5\pi(2-3))}{\pi(2-3)} = \frac{\sin(-0.5\pi)}{-\pi} = \frac{-1}{-\pi} = \frac{1}{\pi} \approx \mathbf{0.3183} \\
n = 1 : \quad h_d[1] &= \frac{\sin(0.5\pi(1-3))}{\pi(1-3)} = \frac{\sin(-\pi)}{-2\pi} = \frac{0}{-2\pi} = \mathbf{0.0000} \\
n = 0 : \quad h_d[0] &= \frac{\sin(0.5\pi(0-3))}{\pi(0-3)} = \frac{\sin(-1.5\pi)}{-3\pi} = \frac{-(-1)}{-3\pi} = \frac{1}{-3\pi} = -\frac{1}{3\pi} \approx \mathbf{-0.1061} \quad (12)
\end{aligned}$$

By even symmetry about  $\tau = 3$  ( $h_d[3+k] = h_d[3-k]$ ):

$$h_d[4] = h_d[2] = 0.3183, \quad h_d[5] = h_d[1] = 0.0000, \quad h_d[6] = h_d[0] = -0.1061 \quad (13)$$

**Step 3: Hamming Window Weights**  $w[n]$  The formula for a length-7 Hamming window ( $0 \leq n \leq 6$ ) is:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{6}\right) = 0.54 - 0.46 \cos\left(\frac{\pi n}{3}\right) \quad (14)$$

Evaluating every tap explicitly:

$$\begin{aligned}
n = 3 : \quad w[3] &= 0.54 - 0.46 \cos(\pi) = 0.54 - 0.46(-1) = 0.54 + 0.46 = \mathbf{1.0000} \\
n = 2 : \quad w[2] &= 0.54 - 0.46 \cos\left(\frac{2\pi}{3}\right) = 0.54 - 0.46\left(-\frac{1}{2}\right) = 0.54 + 0.23 = \mathbf{0.7700} \\
n = 1 : \quad w[1] &= 0.54 - 0.46 \cos\left(\frac{\pi}{3}\right) = 0.54 - 0.46\left(\frac{1}{2}\right) = 0.54 - 0.23 = \mathbf{0.3100} \\
n = 0 : \quad w[0] &= 0.54 - 0.46 \cos(0) = 0.54 - 0.46(1) = \mathbf{0.0800} \quad (15)
\end{aligned}$$

By window symmetry:  $w[4] = w[2] = 0.7700$ ,  $w[5] = w[1] = 0.3100$ ,  $w[6] = w[0] = 0.0800$ .

**Step 4: Windowing Multiplication**  $h[n] = h_d[n] \cdot w[n]$  Multiplying corresponding values tap-by-tap:

$$\begin{aligned}
h[3] &= h_d[3] \cdot w[3] = 0.50000 \times 1.0000 = \mathbf{0.5000} \\
h[2] &= h_d[2] \cdot w[2] = 0.31831 \times 0.7700 = \mathbf{0.2451} \\
h[1] &= h_d[1] \cdot w[1] = 0.00000 \times 0.3100 = \mathbf{0.0000} \\
h[0] &= h_d[0] \cdot w[0] = -0.10610 \times 0.0800 = \mathbf{-0.0085} \quad (16)
\end{aligned}$$

The complete impulse response sequence is:

$$\mathbf{h[n] = \{-0.0085, 0.0000, 0.2451, 0.5000, 0.2451, 0.0000, -0.0085\}} \quad (17)$$

**Step 5: System Function  $H(z)$  and Frequency Response Verification** The  $Z$ -transform is:

$$\begin{aligned}
H(z) &= \sum_{n=0}^6 h[n]z^{-n} \\
&= -0.0085 + 0.2451z^{-2} + 0.5000z^{-3} + 0.2451z^{-4} - 0.0085z^{-6} \quad (18)
\end{aligned}$$

On the unit circle  $z = e^{j\omega}$ , factoring the linear phase term  $e^{-j3\omega}$ :

$$\begin{aligned}
H(e^{j\omega}) &= e^{-j3\omega} [h[3] + 2h[2] \cos(\omega) + 2h[1] \cos(2\omega) + 2h[0] \cos(3\omega)] \\
&= e^{-j3\omega} [0.5000 + 2(0.2451) \cos(\omega) + 0 - 2(0.0085) \cos(3\omega)] \\
&= e^{-j3\omega} [0.5000 + 0.4902 \cos(\omega) - 0.0170 \cos(3\omega)] \quad (19)
\end{aligned}$$

We now verify the filter gains at critical frequencies:

- **DC Gain** ( $\omega = 0$ ):  $\cos(0) = 1$ :

$$H(e^{j0}) = 0.5000 + 0.4902(1) - 0.0170(1) = \mathbf{0.9732} \approx 1.0 \quad (\text{ideal passband}) \quad (20)$$

- **Nyquist Gain** ( $\omega = \pi$ ):  $\cos(\pi) = -1, \cos(3\pi) = -1$ :

$$\begin{aligned} H(e^{j\pi}) &= e^{-j3\pi} [0.5000 + 0.4902(-1) - 0.0170(-1)] \\ &= (-1) \cdot [0.5000 - 0.4902 + 0.0170] = (-1) \cdot [0.0268] = -\mathbf{0.0268} \end{aligned} \quad (21)$$

Stopband attenuation at  $\pi$ :  $|H(e^{j\pi})| = 0.0268 \implies 20 \log_{10}(0.0268) \approx -\mathbf{31.44}$  dB, demonstrating strong lowpass attenuation even with only 7 taps.

---

## Solved Exam Example 2: Highpass Filter Design

**Problem:** Design a 5-tap linear-phase FIR highpass filter with cutoff frequency  $\omega_c = 0.5\pi$  rad/sample using a **Rectangular window**. Determine  $h[n]$  and verify its frequency response at  $\omega = 0$  and  $\omega = \pi$ .

**Solution:**

**Step 1: Symmetry Delay Parameter** Filter length  $N = 5 \implies \tau = \frac{5-1}{2} = 2$ .

**Step 2: Ideal Shifted Highpass Impulse Response**  $h_d[n]$  From Table 2, the shifted ideal highpass formula is:

$$h_d[n] = \begin{cases} 1 - \frac{\omega_c}{\pi} = 1 - \frac{0.5\pi}{\pi} = 0.5000, & n = \tau = 2 \\ -\frac{\sin(0.5\pi(n-2))}{\pi(n-2)}, & n \neq 2 \end{cases} \quad (22)$$

Evaluating each tap explicitly:

$$\begin{aligned} n = 2 : \quad h_d[2] &= 1 - 0.5 = \mathbf{0.5000} \\ n = 1 : \quad h_d[1] &= -\frac{\sin(0.5\pi(1-2))}{\pi(1-2)} = -\frac{\sin(-0.5\pi)}{-\pi} = -\frac{-1}{-\pi} = -\frac{1}{\pi} \approx -\mathbf{0.3183} \\ n = 0 : \quad h_d[0] &= -\frac{\sin(0.5\pi(0-2))}{\pi(0-2)} = -\frac{\sin(-\pi)}{-2\pi} = -\frac{0}{-2\pi} = \mathbf{0.0000} \end{aligned} \quad (23)$$

By even symmetry about  $\tau = 2$ :  $h_d[3] = h_d[1] = -0.3183$  and  $h_d[4] = h_d[0] = 0.0000$ .

**Step 3: Rectangular Window & Multiplication** For a rectangular window,  $w[n] = 1$  for  $0 \leq n \leq 4$ . Therefore  $h[n] = h_d[n]$ :

$$\mathbf{h[n] = \{0.0000, -0.3183, 0.5000, -0.3183, 0.0000\}} \quad (24)$$

↑

**Step 4: Frequency Response Verification**

$$\begin{aligned} H(e^{j\omega}) &= e^{-j2\omega} [h[2] + 2h[1] \cos(\omega) + 2h[0] \cos(2\omega)] \\ &= e^{-j2\omega} [0.5000 - 2(0.3183) \cos(\omega)] = e^{-j2\omega} [0.5000 - 0.6366 \cos(\omega)] \end{aligned} \quad (25)$$

- **Stopband at DC** ( $\omega = 0$ ):  $H(e^{j0}) = 0.5000 - 0.6366(1) = -\mathbf{0.1366} \approx 0$  (attenuated).
  - **Passband at Nyquist** ( $\omega = \pi$ ):  $H(e^{j\pi}) = e^{-j2\pi} [0.5000 - 0.6366(-1)] = 1 \cdot [0.5000 + 0.6366] = \mathbf{1.1366} \approx 1$  (passed).
- 

## 6 Essential Exam Checklist & Traps

- ☐ **Center Tap Formula:** For LPF,  $h_d[\tau] = \omega_c/\pi$ . Do NOT attempt  $\frac{\sin(0)}{0}$  without taking the limit!
- ☐ **Highpass Inversion:** For HPF,  $h_d[\tau] = 1 - \omega_c/\pi$ . All other taps are the negative of the LPF taps.
- ☐ **Window Selection Rule:** If the specification requires  $A_s = 50$  dB, a Hann window ( $A_s = 44$  dB) is insufficient. You **must select Hamming** ( $A_s = 53$  dB).
- ☐ **Argument in Window Cosine:** Remember the denominator is  $N - 1$ , not  $N$ :  $\cos\left(\frac{2\pi n}{N-1}\right)$ .